

Improving Evidence Use and Semantic Verification in Multi-Agent Business Analytics

A Failure-Guided Controlled Study

Mikhail Dushkin

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上海交通大學
SHANGHAI JIAO TONG UNIVERSITY

安泰经管学院
ANTAI COLLEGE
Economics · Management

Master of International Business
Supervisor: Prof. Liu Jialu

Background

Business Calculation Correctness

Average completed sales value per purchasing customer

Customer	Sales value	Status
A	100	Completed
B	60	Completed
C	20	Cancelled

Business definition

Include completed sales only.
Count customers with a completed sale.

All rows: 180 / 3 = 60
Applicable rule: 160 / 2 = 80

Illustrative example. Both calculations execute, but answer different questions.

Background

Two Problems and Two Proposed Mechanisms

1. A past procedure is reused under the wrong business conditions.

   Evidence-checked experience reuse

2. A reviewer approves an incorrect business interpretation.

   Source-grounded counterexample review

Research Question

Research Objective

How do evidence-checked experience reuse and source-grounded counterexample review, separately and together, affect the reliability and cost-effectiveness of a hierarchical multi-agent system for business analytics?

Targeted failures and final-answer quality
Added value relative to strong single-agent controls

Literature Review

DABstep and Existing Analytics Systems

DABstep evaluates multi-step analysis using tables and supporting documents.
Its tasks make business-rule interpretation relevant to final-answer accuracy.

System	Documented approach	Why it matters here
DataPilot	Scoped, versioned assets	Knowledge and procedure reuse
DS-STAR	Planning and iterative review	Review and refinement
NVIDIA	Prepared single agent at inference	Strong architecture comparator

DABstep; DataPilot; DS-STAR v4; NVIDIA Data Explorer · leaderboard context audited 14 Oct 2026

Literature Review

MAST and the Two Failure Problems

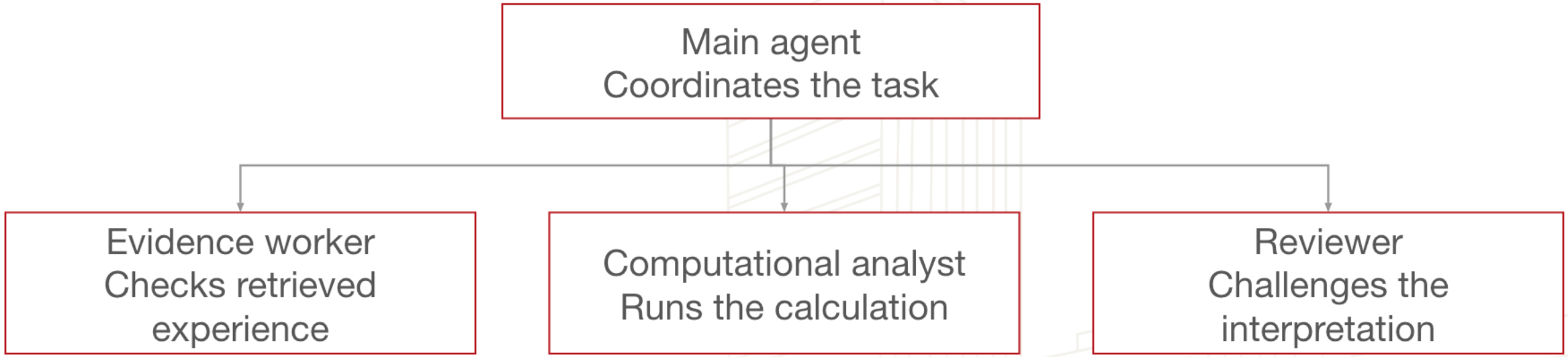
MAST: Multi-Agent System Failure Taxonomy
System design, inter-agent misalignment and task verification

Failure problem	Literature foundation	Question for our adaptation
Experience applied outside its scope	DataPilot assets; ReasoningBank memory	Does current-source checking help?
Required check omitted (FM-3.2) Wrong judgement (FM-3.3)	DS-STAR review; SQLDriller / TS-SQL tests; QueryProof rule gates	Does a distinguishing test improve review decisions?

MAST v3; ReasoningBank v2; SQLDriller; TS-SQL; QueryProof v1

Research Plan & Method

Hierarchical Multi-Agent System

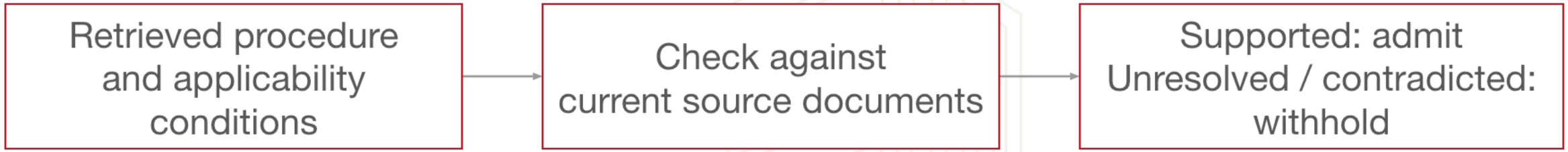


Distinct child contexts and observable transfers
One public-engine adaptation, with the same hierarchy in all MAS conditions

FinHelm; Scaling Agent Systems v3; Strong Single Agent Baseline

Research Plan & Method

Evidence-checked experience reuse



Example: the old procedure includes all orders.
The current definition requires completed sales only.

Measure inappropriate reuse and retention of useful experience.

DataPilot; ReasoningBank v2

Research Plan & Method

Source-grounded counterexample review

1. Read the question and sources before the candidate solution.
2. Construct a small case that separates competing interpretations.
3. Judge the candidate using source-defined expected behaviour.

Interpretation	Result on the illustrative sales case
All orders and all customers	60
Completed sales and their customers	80

Source-defined expectations determine correctness.

SQLDriller; TS-SQL; QueryProof v1

Research Plan & Method

Controlled Experiment and Evaluation

Condition	Experience check	Counterexample review
Baseline	Off	Off
Evidence-checked experience reuse	On	Off
Source-grounded counterexample review	Off	On
Both	On	On

Strong single-agent controls. Same tasks, model, sources and aggregate caps.

DABstep: final correctness and coverage
Same candidate solutions: reviewer errors
Costs: runtime spending, latency and preparation effort

Expected Results & Innovation Points

Expected Research Outputs

Auditable diagnosis of evidence-use and verification failures

Reproducible mechanisms with separate and joint effect estimates

Evidence on reliability and cost for analytical teams

Potential MAST refinements depend on recurring trace evidence.
Positive, null and negative findings all inform the contribution.

Challenges & Limitations

Unresolved Risks and Scope

Challenge	Planned response
Reviewer shares the wrong interpretation or generates an invalid test	Audit test validity and source support
Few relevant failures or poor cost–benefit	Check prevalence in development; report null results
One hierarchy and benchmark setting	Bound conclusions to tested models and tasks

Public data only. No employer data, human study or production deployment.

Expected Research Progress

Prepared Work and Research Schedule

Prepared: literature synthesis, failure map, design, source audit and proposal

Period	Planned work and output
Late October	Minimal harness and agreed model/budget
November	Implement, validate, pilot and freeze protocol
December	Main comparisons and midterm report
January	Failure analysis, uncertainty and cost findings
February–April	Thesis writing, revision and submission
May	Final defense

Which checks improve business correctness enough to justify their cost?